

PHC Pulse — Product Requirements Document

Filled · WHAT are we building?

Purpose. Define WHAT PHC Pulse must do. Filled before handing it to a coding agent.

Filled from the repository PHC-Pulse-GDG-Code-for-community (main @ fd42d41) on 26 Sep 2026. Numbers quoted were measured by running the engine and the 13-test pytest suite (all passing) with PHC_TODAY=2026-09-25, seed 42.

Status legend: **[IMPLEMENTED]** in the repo and tested **[PARTIAL]** partly built **[PLANNED]** designed here, not yet in code

1. Product Summary

PHC Pulse is a supply-chain early-warning and redistribution system for India's Primary Health Centres (PHCs). Every PHC files a one-minute daily report in the worker's own language by web voice note, USSD (*123#), SMS shortcode or an IVR call. Gemini turns the audio or text into a validated, structured report: drug stock, OPD footfall, fever, diarrhoea and respiratory counts, beds and staff. A per-PHC, per-drug forecaster flags stock-outs that will happen before the next monthly indent. A footfall anomaly detector finds outbreak clusters and raises the forecasts for linked drugs. A mixed-integer optimiser then recommends which PHC should ship what to whom, at what cost and ETA. District, state and national officers approve with one tap and track each transfer to delivery on role-scoped dashboards with Gemini briefings. The prototype covers 120 PHCs in 8 districts of 4 states (TN, KA, UP, OD) with 180 days of synthetic, HMIS-shaped history.

Baseline pipeline (as in the submitted deck, now implemented): frontline reporting → AI extraction → forecasting → redistribution → dashboard.

2. Problem

PHCs run out of essential medicines (paracetamol, ORS/zinc, antibiotics, anti-snake venom, oxytocin, malaria ACT) while a PHC 40 km away holds months of idle surplus of the same drug. The cause is how supply is planned. Indents are monthly and based on last month's consumption, supply is uneven, and HMIS data reaches districts weeks late as monthly aggregates. Nobody sees a stock-out coming until the shelf is empty. Frontline workers (ASHA/ANM/PHC staff) have little time, often only a feature phone and patchy connectivity, and are most fluent in a local language, so detailed digital forms go unfilled. Outbreaks such as post-flood diarrhoea or dengue/AES fevers first show up as footfall spikes at a few neighbouring PHCs. These spikes also go unnoticed until IDSP line lists arrive, by which time the linked drugs are exhausted. Officers need daily, forward-looking, actionable signals, not backward-looking monthly reports.

3. Users

User	Need	Permissions
ASHA / ANM / PHC staff	Report today's stock, OPD, syndromic counts, beds and staff in under 60 s, in their own language, on any phone (smartphone, feature phone, landline IVR). Hear or see a read-back to confirm the numbers.	Submit reports for their own PHC only (caller-ID lookup in <i>workers</i> , or PHC persona). Read their own PHC. Cannot approve transfers (API returns 403).
District officer (DHO)	See which PHCs will stock out before the next supply, which neighbours have surplus, and outbreak clusters. Approve transfers, track dispatch and delivery, and send the IDSP alert.	Row-level scope = own district (HMAC-signed token). Approve a recommendation when the donor or recipient is in scope. Update transfer status. Query filters can narrow scope but never widen it.
State / national officer	Command-centre view across districts and states, resilience scores, district roll-up, daily AI briefing in their language, copilot Q&A, HMIS export.	State scope (<i>state-XX</i>) or all rows (<i>national</i>). Same approval rights inside scope. National can enable cross-state lanes in the optimiser.
Admin	Configure Gemini (AI Studio key or Vertex AI), secrets and deployment. Reset demo data between runs.	Environment variables, Secret Manager, Cloud Run. <i>POST /api/admin/reset</i> re-seeds the data. Not role-gated yet: add a role check before the pilot.

4. Goals / Non-goals

Goals	Non-goals
Flag drug stock-outs ≥ 14 days ahead, and only when they land before the next scheduled indent (backtest: 81% recall).	Patient-level records, EMR or prescriptions. Only facility-day aggregates are stored.

Goals	Non-goals
Close shortages by moving idle surplus between nearby PHCs at minimum trip cost, with one-tap human approval and delivery tracking.	Replacing state procurement / DVDMS / eAushadhi or the monthly indent. Unmet need is escalated to the district warehouse, not bypassed.
Detect outbreak footfall clusters within days and pre-position linked drugs (ORS/zinc, PCM/ACT).	Autonomous action. The system never dispatches stock or issues an IDSP alert without an officer.
Daily reporting in < 60 s in 8 Indian languages on any phone type.	Clinical diagnosis or treatment advice to patients.

5. Features

Feature	Description	Priority
Voice / USSD / IVR	[IMPLEMENTED] Web voice note, max 60 s, sent to Gemini as 16 kHz WAV. Returns transcript, English translation, schema-validated report, read-back in the worker's language, and one follow-up question for missing items. USSD *123# menu (stateless replay). SMS grammar <i>PCM 120 ORS 40 BED 4 STF 9 OPD 85</i> . Dialogflow CX webhook for IVR. Languages: hi, ta, kn, or, te, bn, mr, en.	P0
Stockout forecasting	[IMPLEMENTED] Damped Holt-Winters with weekly seasonality on 1,200 PHC × drug series. 28-day horizon, outbreak surge adjustment, risk tiers critical / high / watch / stocked-out, P(stock-out before next supply).	P0
Redistribution optimizer	[IMPLEMENTED] HiGS MILP. Minimises trip cost + handling + urgency- and criticality-weighted shortage penalty. Enforces donor safety stock, consolidates several drugs on one lane, solves per state (cross-state is a toggle), escalates unmet need.	P0
Expiry rescue	[PLANNED] Not in the code: the ledger has no batch or expiry data yet. Design: a <i>stock_batches</i> table (batch, expiry_date, qty). Lots expiring within 90 days whose forecast use falls short become priority donors. Recipients are chosen where forecast use before expiry ≥ quantity. Recommendations are labelled "expiry rescue".	P1
Demand anomaly response	[IMPLEMENTED] Robust weekday-adjusted z-score on fever, diarrhoea, respiratory and OPD counts. Anomalies at ≥ 2 PHCs within 50 km form a cluster. Linked drug forecasts are raised automatically (surge ×1–3, held 14 days). Gemini drafts the IDSP alert in English and the local language.	P1
Capacity / referral	[PARTIAL] Beds occupied and staff present are captured daily and feed the resilience score (−8 if occupancy ≥ 90% or attendance < 60%). Referral routing to the nearest PHC or CHC with free beds is designed but not built.	P1
Dashboard	[IMPLEMENTED] Command centre (KPIs, Leaflet map, district roll-up, resilience score), stock early warning, redistribution, outbreak signals, PHC detail, models & HMIS pages. Gemini briefing and <i>Ask Pulse</i> copilot. Row-level scoped.	P0

6. User Journeys

Journey A: report → structured data

- The ANM at PHC-22 Vandavasi opens *Voice report* and speaks for about 20 s in Tamil: "Paracetamol 150 strips, ORS 60, 3 beds occupied, 9 staff, 72 OPD".
- The browser records and converts to 16 kHz WAV. *POST /api/reports/voice* sends it to Gemini, which returns the transcript, translation and a *ParsedReport* (response schema). Unknown drug codes are dropped server-side.
- The UI shows the fields, the confidence and the Tamil read-back. If an item is missing, Gemini asks one follow-up question and the answer is merged into the report.
- The worker confirms. *POST /api/reports/submit* writes the stock ledger and footfall rows, logs the report (channel, engine, transcript) and invalidates the engine cache. The forecast for PHC-22 updates immediately.
- Feature phones: the same report goes through USSD *123#, SMS or an IVR call (Dialogflow CX webhook), which hear or read back a confirmation.

Journey B: stockout → redistribution

- The engine projects PHC-22 paracetamol at 174 strips, 21.3/day forecast: stock-out in 9 days, next supply in 20. Status is **high**, P(stock-out) = 1.0. PHC-14 Anandapuram holds 5,405 strips (3,038 surplus after safety stock).
- The MILP recommends **REC-C4290993**: PHC-14 → PHC-22, 284 strips paracetamol, 41.5 road-km, INR 1,347, ETA 3.2 h. Reason text: "PHC-14 has surplus paracetamol; PHC-22 needs it in 9 days".
- The DHO Tiruvannamalai clicks **Approve**. A transfer is created (status approved, stock committed at the donor), then **Dispatch** (in transit: stock leaves the donor's ledger), then **Delivered** (stock arrives at the recipient). Forecasts recompute and PHC-22 turns green.

- Any need no donor can meet economically is listed as an emergency indent to the district warehouse.

Journey C: anomaly → response

- Diarrhoea footfall at PHC-91, 93, 95 and 100 (Koraput) reaches up to 4.84× expected for several days. All four are within 50 km, so the engine forms cluster **CL-KRP-diarrhoea** (about 81 excess cases, severity high).
- ORS and zinc forecasts at those PHCs are multiplied by the observed surge. Their stock-out dates move earlier and the optimiser pre-positions ORS/zinc.
- The officer clicks **Draft IDSP alert**. Gemini writes the alert in English and Odia with the likely differential, samples to collect and the logistics. The officer reviews it and sends it. A lone respiratory spike at one Kalaburagi PHC stays a PHC-level anomaly, not a cluster.

7. Success Metrics

Metric	Definition	Target
Stockout lead time	Share of actual stock-out onsets (closing ≤ 0.5 after a positive day) flagged ≥ 14 days earlier, before the next scheduled supply. Walk-forward backtest over the last 60 days.	$\geq 80\%$ recall at $\geq 70\%$ precision. Current: 81.4% recall (302/371), 67.1% precision.
Transfer time	Hours from officer approval to <i>delivered</i> status, from transfer history timestamps. Planned ETA = road-km / 35 km/h + 2 h handling.	Same-district lane ≤ 24 h; median ≤ 12 h. Deck example ETA 3.2 h.
Expiry rescued	Units within 90 days of expiry that are used or transferred before expiry, as a share of all such units. Needs batch/expiry data (planned).	$\geq 50\%$ in pilot; wastage value (INR) reported monthly.
Recommendation acceptance	Approved recommendations ÷ recommendations shown, per week and per officer (transfers table vs plan). Also the share delivered within ETA $\times 1.5$.	$\geq 60\%$ accepted; $\geq 90\%$ of accepted delivered.

8. Constraints

- Connectivity:** feature-phone channels (USSD, SMS, IVR) need no data. USSD is stateless, since the menu is replayed from the path. The web voice path works with or without Gemini (browser speech-to-text plus a rule parser as fallback).
- Privacy:** no patient-level data; only facility-day aggregates keyed by the HMIS NIN. Row-level access by jurisdiction, mirrored by BigQuery row access policies. Audio is not stored, only the transcript and the parsed JSON.
- Human approval:** every transfer needs an officer's approval. Dispatch and delivery are confirmed manually. IDSP alerts are drafts. Every AI output is tagged with its engine.
- Synthetic data:** all history is simulated but HMIS-shaped (NLEM drugs, real district centroids and block names, jittered PHC coordinates, monsoon seasonality, state epidemiology, three injected outbreaks). This is labelled in the UI.
- Deployment:** one Cloud Run container (asia-south1). The prototype uses SQLite inside the container, so it runs as max 1 instance and re-seeds on cold start. Production moves to BigQuery/Firestore. Gemini is configured by key (Secret Manager) or through Vertex AI.